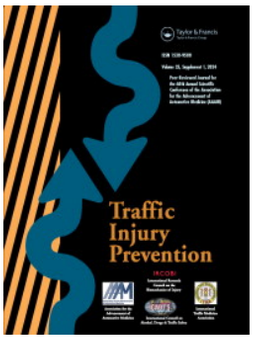

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Baseline vulnerable road user injury risk in multiple U.S. dense urban driving environments

Eamon T. Campolettano[a], John M. Scanlon[a], Ilan Kadar[b], Lev Y. Lavy[b], Daniel C. Moura[b] and Kristofer D. Kusano[a]

[a]Waymo LLC, Mountain View, California; [b]Nexar, New York, New York

ABSTRACT

Objective: Understanding and modeling baseline driving safety risk in dense urban areas represents a crucial starting point for automated driving system (ADS) safety impact analysis. The purpose of this study was to leverage naturalistic vulnerable road user (VRU) collision data to quantify collision rates, crash severity, and injury risk distributions in the absence of objective injury outcome data.

Methods: From over 500 million vehicle miles traveled, a total of 335 collision events involving VRUs were video verified and reconstructed (126 pedestrians, 144 cyclists, and 65 motorcyclists). Data consisted of anonymized video and sensor data (Global Positioning System and accelerometer) from vehicles equipped with Nexar dash cameras. Each event was qualitatively evaluated to assess the collision geometries and other extrinsic collision factors (e.g., time of day, roadway location, presence of occlusions). Previously published injury risk models were utilized to provide estimates of Maximum Abbreviated Injury Scale MAIS2+ and MAIS3+ injury potential. Collision severity distributions and aggregated injury risk estimates were compared.

Results: Vehicle speeds at the time of impact ranged from 0 to 39 mph, and VRU speeds ranged from 0 to 27 mph. In general, vehicles were traveling at lower speeds at the time of impact when turning in comparison to straight travel. Nearly all events (95%) were associated with MAIS3+ injury risk estimates below 10%. Collisions involving a potential occlusion or the vehicle responding to surprising behavior by the VRU were associated with higher estimates of injury risk than those without occlusion or with the vehicle initiating the conflict. Based on accumulated risk for each event, it can be estimated that approximately 55 persons could be moderately injured (MAIS2+) and approximately 6 persons could be seriously injured (MAIS3+).

Conclusion: These data indicate that responding to surprising VRU behavior, having visibility be potentially occluded, and vehicle travel behavior were associated with differences in collision speed and injury risk estimation. The specific comparisons made in this study are not intended to be comprehensive but serve as a starting point for considering baseline driving risk associated with VRU collisions in dense urban areas.

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Crash severity; naturalistic driving data; ADS benchmarking; accumulated injury risk

Introduction

The burden of VRU traffic collisions

Vulnerable road users (VRUs), a group that includes pedestrians, cyclists, and motorcyclists, lack the same crash protections available to vehicle occupants during collision events. The most recent data available from the National Highway Traffic Safety Administration indicate that approximately one-third of all U.S. traffic fatalities in 2021 were VRUs (National Center for Statistics and Analysis 2023a). The number of fatalities represented a record high for each VRU group: 7,388 pedestrians, 966 cyclists, and 5,932 motorcyclists (National Center for Statistics and Analysis 2023a, 2023b, 2023c, 2023d). Though the increase in traffic fatalities across the board is alarming, the magnitude of injured VRUs highlights the high degree of associated harm. According to the Centers for Disease Control and Prevention (2024), in 2021 approximately 116,000 pedestrians, 105,000 cyclists, and 201,000 motorcyclists sought treatment at a hospital following collisions with motor vehicles on public roadways.

Existing data sources

Police-reported crash databases are perhaps the most common source of data used for analyzing automotive safety trends. In the United States, these databases are generally publicly accessible, enabling researchers from numerous disciplines to leverage the most recent data to build upon our knowledge of traffic safety. Given the scale of these databases, which represent billions and/or trillions of driving

CONTACT Eamon T. Campolettano ecampolettano@waymo.com 1600 Amphitheatre Parkway, Mountain View, CA 94043, USA.
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miles, they often serve as the basis for safety impact analysis (Glassbrenner and Starnes 2009; Kusano et al. 2023; Scanlon et al. 2023).

Though police-reported crash data are invaluable, many collisions involving VRUs are not reported to the police or insurance companies (Watson et al. 2015; Shinar et al. 2018; Gildea and Simms 2021). This underreporting phenomenon is consistent with the estimates of VRU injuries being higher based on emergency department records than police reports (National Center for Statistics and Analysis 2023a, 2023b, 2023c, 2023d; Centers for Disease Control and Prevention 2024).

There is a need to better capture "injury-relevant" collisions; that is, those crashes that do not result in injury outcomes but are still associated with meaningful risk of injury. Naturalistic driving studies (NDS), in which vehicles are equipped and monitored for safety events, provide one unique solution for capturing the full spectrum of injury-relevant crashes. The Nexar network, which contributes 200 million miles of new driving monthly in the United States, offers an unprecedented opportunity to capture a wide range of driving events and environments. More information is available in the online supplement.

Traffic safety applications

A large magnitude of injury data available in police-reported crash data is often used in the creation of injury risk curves that relate various factors to injury outcomes. Unlike injury outcomes, which are dependent on a multitude of factors, injury risk curves provide a means for contextualizing a given collision event's propensity to create injury harm. VRU injury risk curves often use some measure of vehicle speed, as well as other vehicle, person, and biomechanical factors as predictor variables. They have been developed for vehicle occupants, as well as for VRUs (Rosen 2013; Ding et al. 2019; Lubbe et al. 2022; Schubert et al. 2023, 2024). It should also be noted that leveraging objective injury outcomes in a predictive manner through the development of injury risk curves underpins the automotive safety industry, most notably in NHTSA's New Car Assessment Program and the European New Car Assessment Programme. Risk curves are commonly used in estimating collision severity for real collisions or in predictive form in the evaluation of simulated collisions (Sander and Lubbe 2018; Riexinger et al. 2019; Scanlon et al. 2021). For instance, NDS data notably lack injury outcomes, but injury risk assessments may still be carried out by leveraging the available data from crashes. Additional references are available in the online supplement.

Another application of crash data is retrospective safety impact, or the historical analysis of a safety system's ability to prevent and mitigate injuries (Glassbrenner and Starnes 2009; Fildes et al. 2015; Cicchino 2017; Flannagan 2023; Kusano et al. 2023). A key requirement of these assessments is a benchmark against the current driving population from which to measure system effectiveness. Most prior automated driving system (ADS) benchmarking studies have relied on police-reported crash data to evaluate performance (Blanco et al. 2016; Favarò et al. 2017; Teoh and Kidd 2017). However, as previously noted, these data are subject to underreporting and systematically miss injury-relevant crashes where an injury does not occur. As an alternative, this study investigates the potential of a large-scale naturalistic driving study (Nexar) for creating performance benchmarks using the concept of "injury risk accumulation," whereby events are measured by the propensity to cause injury, rather than the occurrence of injury itself, and the accumulation of this risk is measured over some amount of driving exposure.

Research questions and study objectives

The purpose of this study was to leverage naturalistic VRU collision data (Nexar) in conjunction with injury risk models to quantify baseline collision rates, crash severity, and injury risk distributions in the absence of objective injury outcome data. Two primary research questions were posed: (1) How does the VRU collision rate vary by dense urban environment? and (2) What extrinsic factors are associated with differences in collision severity? Additionally, metrics for summarizing aggregate injury risk estimates are presented to assess relative crash severity and to quantify the expected injury burden associated with this data sample.

Methods

Data source

The data in this study represent a multiyear sample (November 2018 to February 2023) from vehicles equipped with Nexar dash cameras. Only data consisting of anonymized video and sensor data (Global Positioning Systems and accelerometers) from those vehicles involved in collisions with VRUs were included.

The overall data set consisted of 335 collision events. Approximately 80% of these events occurred in New York City, as identified based on the full list of ZIP codes within the 5 boroughs (Table 1). Improvements in the event recall process were in place for all events from July 2020 onward. As such, example city-level rate comparisons are presented that only consider the collision and mileage data after this time point. The known mileage from these cities totaled over 500 million miles. Mileage data were provided for all cities in which at least 3 collisions were observed. There were 27 events from localities in which either the location was unknown or only a single event occurred; mileage data were not available for these events.

Table 1. Summary of collision data and mileage by geographic location.

Location	Collision count	Collision count (after July 2020)	Mileage after July 2020 (million miles)
New York City	277	165	385.6
Los Angeles	12	12	71.5
San Francisco	6	5	7.6
Chicago	6	4	11.0
Phoenix	4	4	25.1
Las Vegas	3	2	10.3

Collision identification and reconstruction

All collision events were identified by Nexar using a combination of kinematic triggers, video review, and individual reporting. Identified collision events were reconstructed using a combination of video, sensor, and mapping data. Specifically, vehicle sensor data (Global Positioning Systems and accelerometers) and video were used to calculate vehicle speed at impact, and the video data, in conjunction with spatial information, were used to estimate VRU agent speed and impact time/location. Detailed information regarding the collision identification and event reconstruction routines is available in the online supplement (Kadar et al. 2022).

Video review

Video for each collision event was reviewed to visually verify collisions and evaluate several qualitative features of the collision. This review process involved assessing the collision configuration and impact location to inform the injury risk assessment. Impacts were identified as being either frontal or side contacts. Contacts with the side of the vehicle forward of the vehicle A-pillar, approximated based on the field of view of the dash camera, were considered as being frontal contacts. This designation captures the fact that the physical engagement for VRUs in this configuration is often comparable to a purely frontal contact (i.e., that the VRU engages with the hood and may engage with the A-pillar and/or windshield) and is consistent with previously published injury risk curves (Simms and Woods 2009; Schubert et al. 2023, 2024).

A simplified conflict type grouping was utilized in the present study. Specifically, each case was assessed based on the vehicle's travel behavior (straight, turning left, or turning right) and whether the collision occurred at an intersection or not. For each case, the vehicle's role as either the initiator or responder in the collision was also considered, where the initiator role corresponds to events in which the vehicle performed an initial, surprising behavior that led to the conflict (Scanlon et al. 2021; Dinparastdjadid et al. 2023). The responder role corresponds to events in which a conflict occurred due to some initial surprising behavior of the involved VRU. For example, an event in which a cyclist attempted to cross an intersection during a red light signal phase and was struck by a vehicle traveling through the intersection on a green light would be evaluated as a responder.

Lastly, the presence of geometric occlusions (e.g., other vehicles, buildings, vegetation) potentially limiting the vehicle's or VRU's response was noted for each case. Further, for collisions involving cyclists and motorcyclists, VRU helmet status was noted.

Severity assessment

Information from the collision reconstructions, including vehicle speed, VRU speed, and collision configuration, were leveraged to compute injury risk estimates for each collision. The risk curves used in this analysis, unique to pedestrians, cyclists, and motorcyclists, as well as collision configuration considerations, are from the literature and are presented in the online supplement (Lubbe et al. 2022; Neale et al. 2021; Schubert et al. 2023, 2024). As noted in the Introduction, collision injury risk models capture some measure of speed and oftentimes include other predictors. In some cases, these predictor variables may not always be feasible to attain or even be known in practice (McMurry et al. 2021). Evaluating for each of these predictors in full may lead to overindexing on the specific factors associated with the VRU for that given event rather than the generalized nature of the collision. For instance, attempting to apply a risk curve with VRU age as a predictor variable would appropriately result in higher levels of injury risk predicted for an older individual compared to a young adult. Using a consistent age across events instead would allow for consistent comparisons to be made within the data set, as well as with other data sets. This practice of effectively reducing the variables considered using default predictors generalizes the data sample, while mitigating sampling biases associated with the predictors. Accordingly, considerations for developing default predictors are presented in the online supplement, such that all risk evaluations only vary on a measure of collision speed, measured in miles per hour. As such, these event estimates limit some of the available event context to reflect a population-weighted average risk rather than an event-specific risk estimate. Lastly, only a single injury risk model was applied for each VRU type, though the literature contains various models. The authors sought the most recent injury risk models, and it should be noted that the selection of a given model is less important than ensuring that the same model is used when making comparisons across different data sets.

VRU collision metrics

Collision rates

Rates were first computed to capture the collision rate for each geographic area and for crash type to capture the frequency at which contact events occur over a given driving mileage. All collision rates in this study are presented on a per million mile basis. Confidence intervals (95%) for these rates were developed using Byar's method, which represents an exact approximation to the Poisson distribution and retains high levels of accuracy for both small and large counts (Breslow et al. 1980). The lower (Eq. (1)) and upper (Eq. (2)) bounds for the 95% confidence intervals of counts are given by the following equations:

$$L.B. = a\left(1 - \frac{1}{9a} - \frac{1.96}{3}\sqrt{\frac{1}{a}}\right)^3 \tag{1}$$

$$U.B. = (a+1)\left(1 - \frac{1}{9(a+1)} + \frac{1.96}{3}\sqrt{\frac{1}{a+1}}\right)^3, \tag{2}$$

where a is the number of observed events and 1.96 is reflective of the Z-score corresponding to a 95% confidence interval. Subsequently dividing by the relevant mileage results in confidence intervals for the rate data.

Accumulated injury risk

To reflect that not all collision events are equal in severity and that injury potential increases nonlinearly with increasing collision speed, comparative metrics related to injury risk were also computed. Accumulated injury risk (Maximum Abbreviated Injury Scale MAIS2+ and MAIS3+) was defined by summing the individual risk estimates for each collision and then aggregating across all events within a specified grouping (e.g., pedestrian collisions at night or intersection cyclist crashes). Absent objective injury outcome data, this metric reflects the expected number of injuries at the given severity level over the analyzed driving miles. Though the accumulated injury risk values are not integers, they are representative of injury outcome counts, so modeling using the Poisson distribution to compute 95% confidence intervals was deemed appropriate.

Injury risk distributions

The injury risk distributions at each severity level were quantified (median, 75th and 95th percentile injury risk values) to allow for relative event severity comparisons among extrinsic collision factors. The data were resampled with replacement ($n = 1{,}000$) to generate 95% confidence intervals for these point estimates. The lower and upper bounds were obtained by sorting the samples and taking the 25th and 975th values, respectively. Resampling was done within each grouping to ensure fidelity of the bootstrap samples to the original data set.

Results

Descriptive statistics

A total of 335 collision events were video verified and reconstructed, of which 126 involved pedestrians, 144 involved cyclists, and the remaining 65 involved motorcyclists. Mileage data totaling 510 million driving miles were available for 193 of these collision events.

Video for each collision event was reviewed to quantify several extrinsic features hypothesized to relate to collision severity and injury risk: Conflict type (combination of roadway location and vehicle turning behavior), time of day, presence of occlusions, and conflict role. As shown in Table 2, more than 75% of all collisions occurred at roadway intersections. Similarly, most events involved the vehicle traveling straight at the time of impact (69%). More than half of all pedestrian and motorcyclist events were observed to occur at night, whereas approximately 60% of cyclist events occurred during the day. Conflict role was observed to vary considerably by VRU type, with vehicles initiating more than double the number of conflicts they responded to for pedestrians, while the inverse relationship was observed for collisions involving cyclists (Table 2). VRU failure to yield to a moving vehicle represented the most common VRU initiating behavior, occurring in more than half of all vehicle responder events. This is representative of VRUs continuing to travel through intersections or crosswalks when the vehicle had the right of way and was in motion. Notably, improper lane keeping by cyclists was observed in approximately 25% of cyclist initiator events. From a vehicle initiator standpoint, the most common behavior was failure by the vehicle from a stopped position to yield to the VRU (e.g., attempting to complete a turn from a stopped position with a pedestrian in the crosswalk). Occlusions potentially limiting the agent's ability to recognize the other involved party were observed to be present in fewer than one-third of all events. Lastly, helmet usage was observed for 31% of cyclists (44/144) and 57% of motorcyclists (37/65). It must be noted that driving exposure to these various features (e.g., amount of time driven at night versus day) was not explicitly explored and therefore the relative collision risk with respect to these features was not evaluated.

Table 2. Summary of qualitative features of collision events as identified by video review.

	Cyclist, *n* (%)	Motorcyclist, *n* (%)	Pedestrian, *n* (%)
Roadway location			
Intersection	113 (78)	47 (72)	99 (79)
Not at intersection	31 (22)	18 (28)	27 (21)
Time of day			
Day	87 (60)	27 (42)	57 (45)
Night	57 (40)	38 (58)	69 (55)
Vehicle turning			
Straight	108 (75)	52 (80)	72 (57)
Left turn	25 (17)	12 (18)	42 (33)
Right turn	11 (8)	1 (2)	12 (10)
Vehicle conflict role			
Initiator	50 (35)	31 (48)	87 (69)
Responder	94 (65)	34 (52)	39 (31)
Geometric occlusion			
No occlusion	99 (69)	44 (68)	104 (83)
Potential occlusion	45 (31)	21 (32)	22 (17)

All values in the table represent collision counts in each grouping with percentages included in parentheses.

Based on the reconstructions, at the time of impact, vehicle speeds ranged from 0 to 39 mph (Figure 1). For VRUs, speeds ranged from 0 to 13 mph for pedestrians, 0 to 16 mph for cyclists, and 0 to 27 mph for motorcyclists (Figure 1). When considering these agent speeds in conjunction with the collision configuration, the collision speed input used in the injury risk models as defined in the online supplement was observed to vary from 0 to 39 mph.

In aggregate, the events in this data set were observed to be associated with a wide range of moderate or greater (MAIS2+) injury potential, with low-end values approaching 0% risk and an upper end risk over 60% (Figure 2). Intuitively, the range of injury risk estimates was narrower at the MAIS3+ level, with more than 95% of all events associated with an MAIS3+ injury risk less than 10%. Though the distribution of collision speeds in pedestrian events was observed to be right-skewed relative to both cyclists and motorcyclists, the distribution of injury risk estimates is higher for pedestrians for MAIS2+ injury risk (Figure 2). Motorcyclist collision events were associated with a higher proportion of MAIS3+ injury risk than either cyclists or pedestrians.

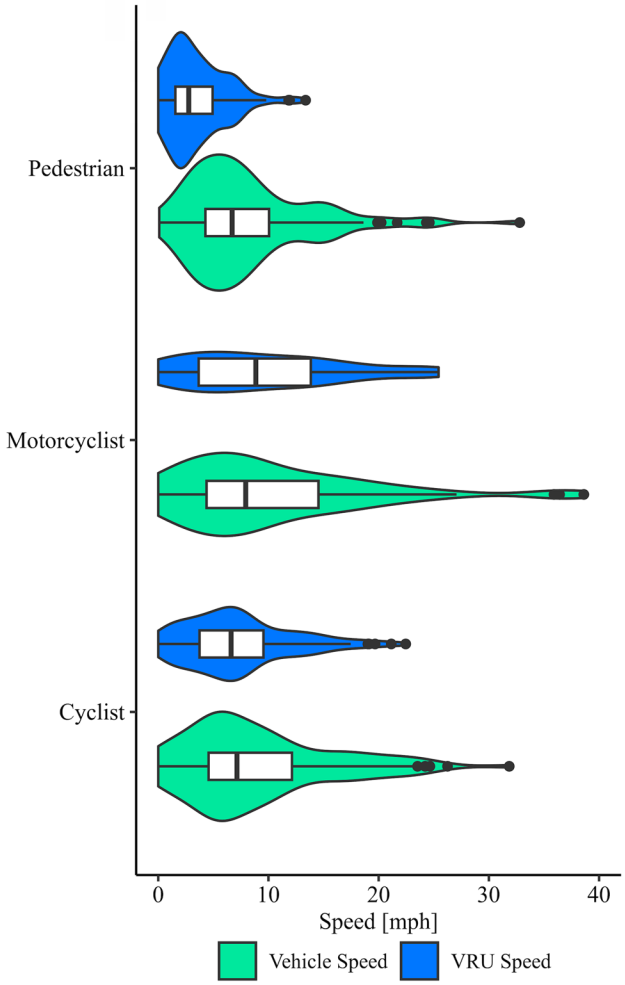

Figure 1. Speed distributions at the time of impact by VRU type.

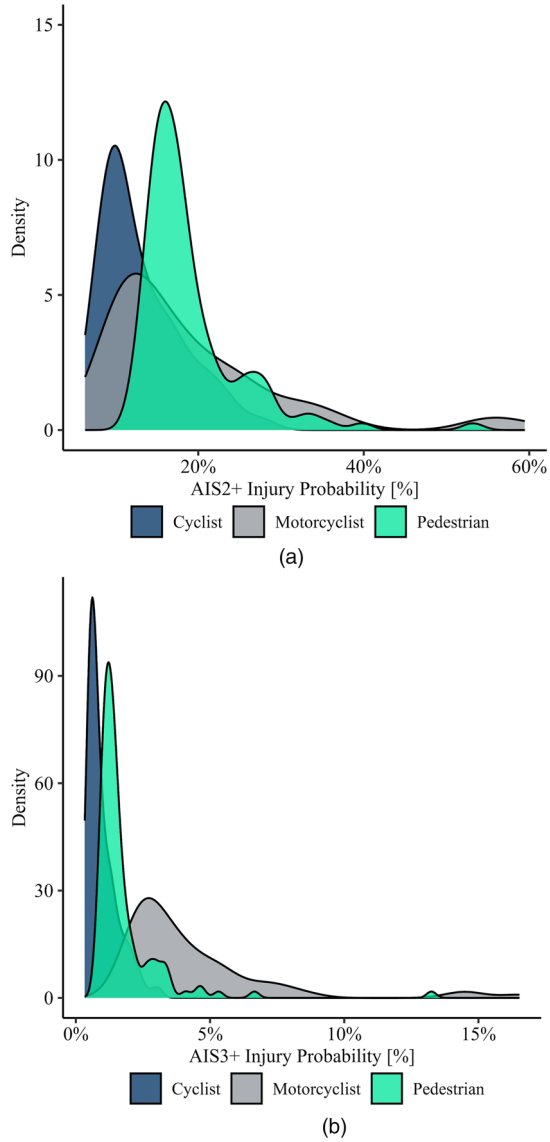

Figure 2. Distribution of injury risk potential at the MAIS2+ (a) and MAIS3+ (b) levels by VRU type.

Comparisons based on the extrinsic collision features assessed for each case (see Table 2) were also carried out without stratification across geographic location given the composition of the data sample and are available in the online supplement.

Accumulated injury risk

Individual risk estimates from each collision event in the data set were summed to compute an aggregate measure of expected injury burden. The accumulated MAIS2+ injury risk from all events in this data set totaled 55.1, and the accumulated MAIS3+ injury risk totaled 6.4. These values represent the expected total number of injuries of moderate or greater (MAIS2+) severity and serious or greater (MAIS3+) severity that could have occurred in the 335 VRU collisions in the present study.

Geographic-based collision rates

Though driving mileage and collision events from New York City dominate this data set (Table 1), aggregate comparisons of collision rates were computed, with 95% confidence intervals capturing the inherent uncertainty associated with more limited data collection in certain geographic areas. San Francisco was associated with the highest rate of collisions of all geographic areas considered, though the confidence intervals overlapped with all other cities given the low number of collisions observed there (Figure 3a, top). Despite having a collision rate approximately equal to that of Los Angeles, Phoenix had approximately double the rate of expected moderate or greater severity (AIS 2+) injuries (Figure 3b, bottom).

Extrinsic crash factors influencing severity

Given the high event counts for straight intersection collisions for all VRU types, accumulated MAIS2+ and MAIS3+ injury risk is highest for this collision configuration (Figure 4). This means that for these turning and intersection combinations analyzed, collision events with non-turning vehicles at intersections represented the highest amount of accumulated injury risk. It should be noted that the accumulated injury risk metric is dependent on the number of events in a given grouping and a higher value may not necessarily be

reflective of heightened individual event severity. To appropriately decouple the frequency and severity components, normalization by event count may be useful for specific comparisons regarding relative event severity. Given the sparsity of data for some of the turning intersection combinations, a normalized comparison was not carried out for these data.

Event severity distributions were quantified and compared within VRU type for each of the extrinsic collision factors considered in this analysis: Time of day, conflict role, occlusion, vehicle turning, and roadway location. This analysis demonstrates the severity differences given that some event occurred (i.e., controlling for the rate at which crashes with some collision factor occurred). Comparisons were not made between VRU types given the differences in injury risk models used for each type, which could confound any conclusions. Comparisons at the MAIS2+ severity level are presented in Figure 5, and the MAIS3+ comparisons are available in Figure A4 (online supplement). Though not statistically significant, the 95th percentile risk values for MAIS2+ and MAIS3+ injury risk for pedestrian events are higher for collisions at night compared to the day (Figure 5). Events with a potential occlusion were associated with higher median and 75th percentile injury risk at both the MAIS2+ and MAIS3+ severity levels for pedestrians, which suggests that the collision events with occlusions present tend to be higher severity (Figure 5). Collisions with pedestrians in which the vehicle was in the responder role were associated with increased risk at both the MAIS2+ and MAIS3+ levels (Figure 5). There were no observed differences in severity distributions based on whether the collision occurred at/near the intersection versus not at an intersection. Collisions involving a straight-traveling vehicle had higher 95th percentile MAIS2+ and MAIS3+ injury risk than those involving turning vehicles. For pedestrians, the distributions were similarly higher severity for collisions with straight-traveling vehicles, though beginning at the 75th percentile.

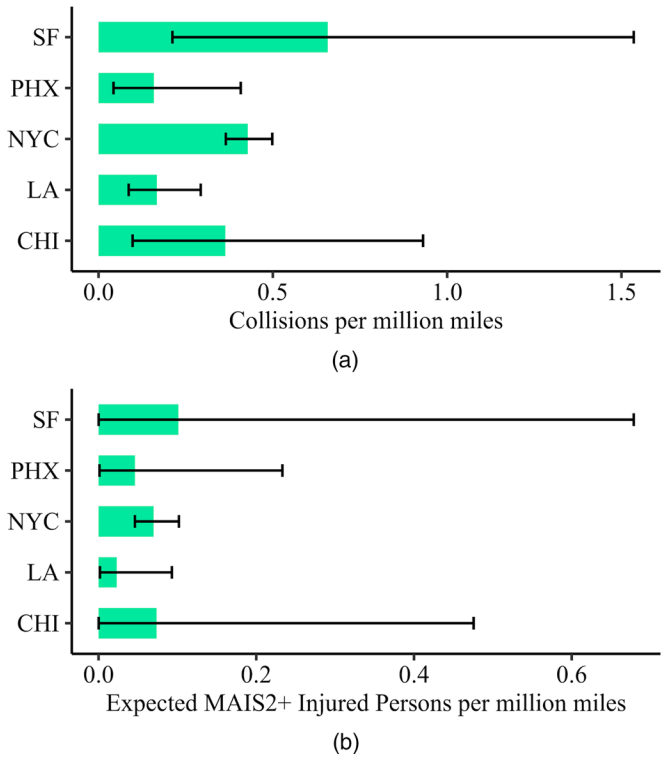

Figure 3. (a) collision rate and (b) accumulated moderate injury risk (per million miles) by geographic region. Bounds reflect 95% confidence intervals on the observed collision counts defined as a Poisson process as described in the Methods.

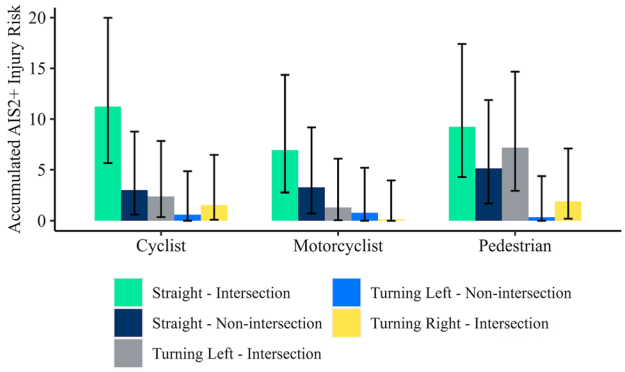

Figure 4. Accumulated MAIS2+ injury risk for each VRU type based on conflict grouping. The confidence intervals were calculated based on the expected injury "counts" as described in the Methods.

Discussion

From over 500 million driving miles, collision reconstructions and severity distributions from 335 events involving VRUs were presented. To our knowledge, this analysis represents the largest sample of NDS data from collisions involving VRUs in the United States. As a point of comparison, the second Strategic Highway Research Program, which is the largest organized NDS in the world to date and which collected approximately 32 million miles of driving from over 3,500 drivers in 6 geographic locations over a 4-year period, captured fewer than 10 collision events involving VRUs (Antin et al. 2019). The specific comparisons made in this study are not intended to be comprehensive; rather, they serve as a starting point for considering baseline driving risk associated with VRU collisions in dense urban areas.

Comparisons to other NDS data

This study builds on previous investigations of VRU severity distributions for pedestrians and cyclists based on dash camera data from taxi drivers in Los Angeles (Campolettano et al. 2023; 2024). Elsewhere, dash camera data from taxicabs operating near Seoul, South Korea, have served as the primary source for investigating collision severity and VRU-specific circumstances. Based on 357 vehicle-to-pedestrian collision events, Chung (2018) reported injury severity on an ordinal scale based on police reports and descriptive statistics on vehicle speed at the time of impact (mean = 27.4 kph [17.0 mph], SD = 22.6 kph [14.0 mph]). Not surprisingly given the police report requirement for Chung's study, the average vehicle impact speed in the current study was approximately 50% lower (mean = 14.3 kph [8.9 mph], SD = 11.1 kph [6.9 mph]). This same data source was leveraged most recently by a multi-institution endeavor in Germany, ATTENTION, to facilitate the development of real-time VRU injury risk prediction.

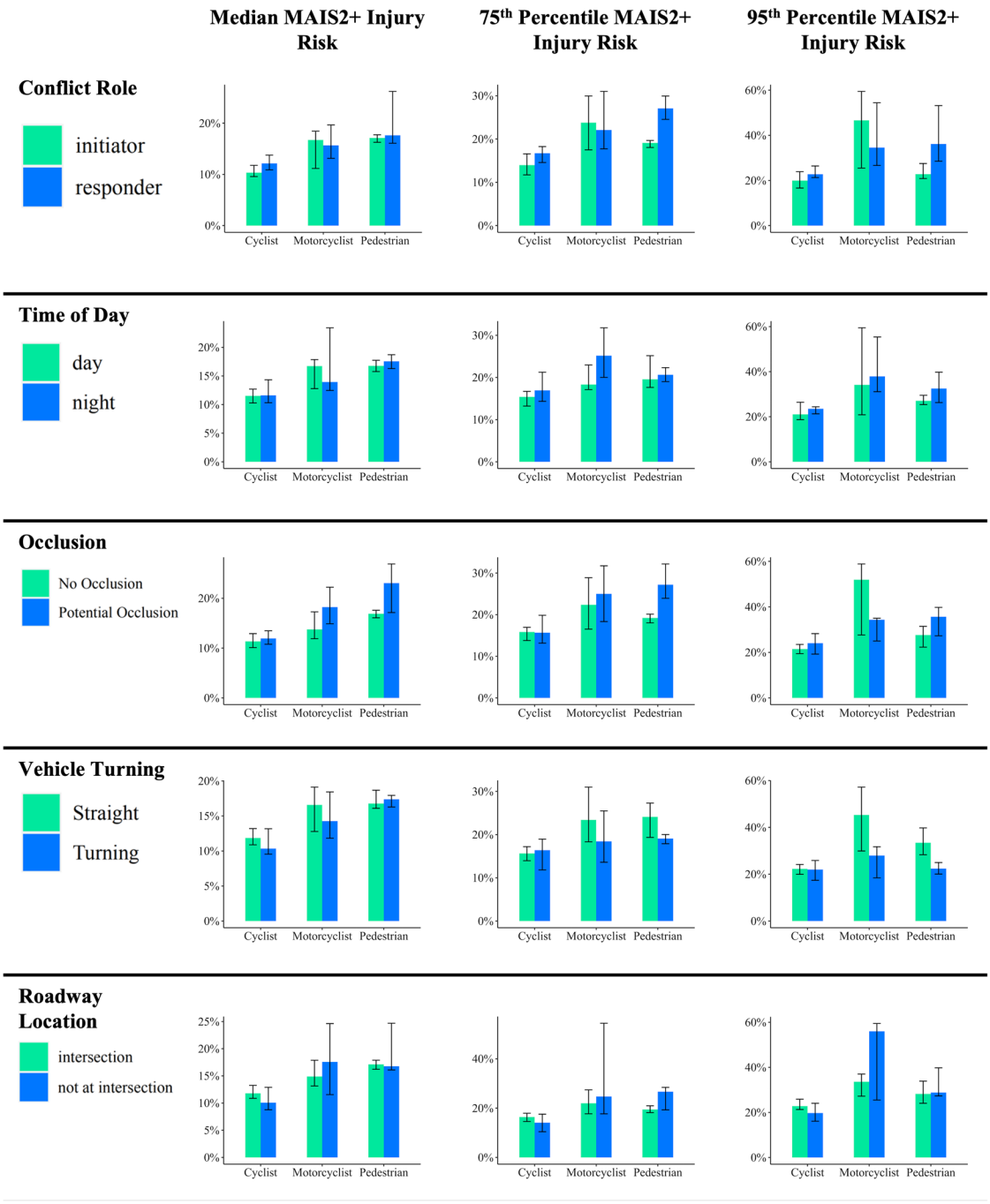

Figure 5. Comparison of event severity distributions at the MAIS2+ severity level for each VRU type by extrinsic collision factors. Confidence intervals (95%) represent bounds from bootstrap analysis of the available collision data as described in the Methods.

Researchers used the dashcam video to characterize VRU pre-collision behaviors and postures for eventual input into crash simulations (Lich et al. 2023). Leveraging NDS data for such purposes highlights the potential to answer a variety of questions from these rich sources of data.

Study implications

These results highlight the importance of considering both frequency and severity in conjunction when evaluating collisions (Kullgren 2008). Looking strictly at frequency will provide insight into how common events may be but may not be indicative of how harmful those events could be. Similarly, only looking at injury risk can identify those configurations associated with the highest injury potential but overlooks how uncommon those events may be. For instance, for pedestrians, there was a 60% increase in left turn at intersection collisions in comparison to straight-traveling non-intersection collision events; however, accumulated risk at the MAIS2+ and MAIS3+ levels was only higher by 39% and 16%, respectively (Table 2 and Figure 4). In absolute terms, average injury risk for

pedestrians in these collisions was approximately 3% higher (20.6% vs. 18.0%) for MAIS2+ injuries and approximately 0.6% higher (2.1% vs. 1.5%) for MAIS3+ injuries. Given the right-skewed nature of the speed and injury risk distributions, utilizing average or median risk for comparisons may mask differences in severity; comparing 75th percentile (3rd quartile) risk or 95th percentile risk may be more appropriate to differentiate distributions with higher severity signal (Figure 5). Further, consideration of extrinsic collision factors, such as those presented here, may have an effect on a given collision event's severity. In this study, responding to surprising VRU behavior, having visibility be potentially occluded, and travel behavior were associated with differences in collision speed and injury risk estimation (Figures 4 and 5). Though time of day differences in collision severity were not observed in this study, exposure (i.e., driving mileage) was not bifurcated to allow for a comparison in collision rates.

The presented analysis offers distinct advantages to other publicly available data in the United States. Firstly, the United States has limited VRU crash injury data collection tooling in place compared to what is carried out for vehicle-to-vehicle collisions. Previous in-depth endeavors in the United States, like the Pedestrian Crash Data Study or Motorcycle Crash Causation Study, focused on very small geographic areas that may not be representative and were designed to address specific questions. These data are limited in case counts, and, in the case of the Pedestrian Crash Data Study, which stopped data collection in 1998, may not be relevant to today's VRUs. It must be noted that the collision data in this study were primarily from New York City, and the results may not be generalizable to other urban areas.

Those events that may not result in hospitalization or a police report are captured in the present data set, where they may be missing from other injury surveillance databases, such as Crash Investigation Sampling System (CISS) or the German In Depth Accident Study, based on sampling criteria. Injury risk models were utilized in the absence of objective injury outcome data to estimate the risk of injury for the VRUs in the reconstructed collision events. Injury outcome data are inherently variable. As an example, consider 2 male pedestrians of the same age struck by the front of a vehicle traveling at 20 mph in separate events. One pedestrian was not injured beyond some scrapes associated with a subsequent fall to the ground (AIS 1 injury), whereas the other pedestrian experienced a fractured femur due to direct contact with the vehicle (AIS 3 injury). The first pedestrian would likely not be captured in most police-reported data sets, whereas the second would, despite the fact that the factors related to their collisions were exactly the same. Risk-based metrics employed predictively necessarily allow for applying the same approach to collision events, ensuring that the estimated risk would be equivalent for both the pedestrians in the above example. When extrapolating this example to a data set with several hundred events with a wide variety of involved VRUs, as in the present study, or to simulated collision events with nonspecific VRU agents, the power of a risk-based approach becomes clear.

Preventing serious injuries and fatalities represents the focus of the traffic safety community. Taking a risk-based approach toward safety evaluation allows for the specific collision scenario to be evaluated for its injury potential rather than indexing exclusively on those cases with injury outcomes. Further, a risk-based approach ensures that there is some injury potential that can be estimated for all events. By aggregating all of these injury risk estimates, an estimate of the overall injury burden may be determined. Based on accumulated risk, it can be estimated that approximately 55 persons in this data set could be moderately injured (MAIS2+) and approximately 6 persons could be seriously injured (MAIS3+). This allows for a safety impact lens to be applied to events that otherwise may not be captured by existing surveillance or sampling systems.

Potential applications

Though NDS data often allow for observation of all vehicle contact events, considering them all as collisions is dependent on the research question posed. In the current study, where the goal was to present an application of developing VRU collision baselines, inclusion of all video-verified contact events was deemed appropriate. As it pertains to injury-relevant scenarios, filtering out or assessing relative injury potential of collision data using a speed or estimated injury risk threshold may be appropriate. For example, ISO 26262 uses the "S" severity grading scale using 10% risk of AIS severity level (AIS 1+, AIS 3+, AIS 5+) to bucket events according to injury risk for functional safety applications (International Organization for Standardization 2018).

The foregoing analysis can serve as the starting point for ADS comparisons. Prior to deployment, appropriate and relevant NDS data may be used as a comparison point to simulated driving events from many millions of miles to assess the safety performance of the ADS for a given operational design domain (ODD). Given that injury outcomes are, relatively speaking, rare, statistical assertions based on high-severity injury outcomes exclusively would require an extreme number of miles (e.g., billions of miles to prove a reduction in fatal injury outcomes; Kalra and Paddock 2016; Lindman et al. 2017; Scanlon et al. 2023). Under a risk-based approach, accumulated injury risk at a given severity level, which can be viewed as a correlate for objective injury outcome data, could be compared to risk-based numbers from NDS data or objective injury outcome data to determine to what extent the ADS provides a positive safety impact relative to human drivers.

Given the distribution of this data set (Tables 1 and 2), additional comparisons beyond those presented were not able to be undertaken. For example, if one wanted to evaluate whether the collision rates for motorcyclists in Phoenix and New York were similar, the aggregate results presented here indicate that further considerations, such as time of day and roadway location (intersection vs. non-intersection), may be valuable, provided the mileage data could be bifurcated appropriately to capture any differences in driving mix.

For this specific comparison, it would also be necessary to take into account differences in helmet usage. New York has a law requiring all motorcyclists to wear a helmet when riding, whereas Arizona does not have a law for adult riders. It is well documented that unhelmeted cyclists and motorcyclists experience more serious injury outcomes than their helmeted counterparts (Khor et al. 2017; Scott et al. 2019; Lee et al. 2022). Similarly, cyclist helmet usage has been shown to be lower for public bike shares than for personal bicycles (Fischer et al. 2012). Accordingly, comparisons of cyclist collision and/or injury rates across cities should aim to account for this phenomenon. These examples are illustrative of potential factors related to collision and injury rates. Appropriate consideration regarding what factors are and are not captured in the computed rates should be noted when making any comparisons.

The events in this data set represent a wide range of vehicle and VRU speeds at impact, as well as a wide range of moderate or greater (MAIS2+) injury risk estimates, and provide some signal on low-end serious or greater (MAIS3+) injury risk estimates. Nearly all events were associated with less than 10% MAIS3+ injury risk (Figure 2). Though this represents the vast majority of collision events, higher severity events, which are overrepresented in the crash injury databases, are less represented in the present data set. Thus, to get a complete perspective on the collision space, consideration of both the high-frequency, lower severity crashes (like in this data set) and the lower frequency, higher severity crashes (as captured in crash databases) is crucial. There are several potential avenues to do so, two of which are presented below. One option would be leveraging extreme value theory to fit a distribution to the underlying NDS data to estimate the frequency and severity of the long-tail collision events (Bin-Nun et al. 2024). Alternatively, efforts to join the NDS data sets with crash databases may also be explored. By assuming that the NDS data are representative for the crashes it does capture, the crash database data may then be reweighted to reflect its increased sampling of higher severity events. A cutoff threshold where the NDS data are no longer considered may be appropriate based on impact speed or injury risk at a given severity level. This combined approach allows for a full range of collision severities to be considered and maximizes the utility of NDS data, which can often be quite expensive to collect.

Injury risk model considerations

The data used to generate injury risk curves generally represent observational samples of injury outcomes that are necessarily censored. For injured persons (left-censored data), there is no way of telling how much less severe the collision could have been and still resulted in the same injury severity. Similarly, for uninjured persons (right-censored data), there is no way to know how much more severe the collision could have been and still resulted in no injuries. This double censoring is also reflective of biomechanical variance. Additionally, it can be difficult to differentiate between injuries directly caused by contact with the motor vehicle and those caused by potential subsequent contacts, such as those associated with a fall to the ground.

The sampling criteria for the data sets used to develop VRU injury risk models are biased toward higher severity events that trigger the crash for reporting to the police and/or investigation. This often results in injury risk curves presenting nonzero injury risk at speeds of zero, particularly at lower injury severity levels. Efforts to refit these risk curves to limit the overprediction of low-end risk may be useful. It is important to be aware of these potential biases when working with injury risk models and to ensure that consistent use of a specific risk model is used when attempting to make comparisons across data sets.

This analysis represents the largest, and most complete, sample of NDS data from collisions involving VRUs in the United States. These data indicate that considering both collision counts and severity is crucial to understanding the landscape of crash injury risk. The specific comparisons made in this study are not intended to be comprehensive but serve as a starting point for considering baseline driving risk associated with VRU collisions in dense urban areas.

Limitations

There are several limitations to note. This agent speed-based reconstruction and injury risk assessment methodology applies well to impact events; however, collision events involving runover, crushing, or entrapment are associated with different injury mechanisms that are not well suited to existing risk models. There were no instances in this data set where an impact event was not the initial event, though. Consideration of the relevant injury mechanisms should be considered when presenting collision and/or injury rates for comparisons.

Though this data set represents the largest of its kind in the United States, over 80% of the collision events are from New York City. Accordingly, the non-geographic rates presented herein are biased toward New York City. The large confidence intervals associated with the other geographic areas highlight the limited data sampling in those locations and the associated uncertainty with those presented rates. Also, roadway mileage was reported in aggregate across time of day and roadway type (i.e., highway vs. surface streets), and exposure to these driving miles may actually vary considerably, resulting in very different rates estimates. These results may not be generalizable to other dense urban ODDs or to non-dense urban ODDs. Again, it must be noted that this study serves as one approach toward considering baseline driving safety risk for VRU collision events and is not meant to present the true measure of VRU collision rates for these cities.

There are also notable limitations with the injury risk models employed in this study that may influence the accuracy of any individual risk estimate. These injury models were developed using German crash data from the last 20 years. Differences in the composition of the driving fleet, as well as in the use of protective equipment for cyclists and motorcyclists, relative to the United States may be associated

with unquantified differences in injury risk. Further, only cases with at least a suspected VRU injury were sampled, resulting in data censoring and low-end risk offsets. The published models were adapted and modifications to extend application of the models to other crash modes were made in consideration of available data to ensure their appropriateness. Similarly, the injury risk models use different measures of speed, and a single consistent measure was developed for use in this data set for input (without adjustment) into the published risk curves. Accordingly, these changes may result in estimates of injury risk that may not directly align with the published risk models. These systematic modifications were implemented consistently, though, mitigating their effect in data comparisons.

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